

Reverse Computational Research Hub Group

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Chapter 1

What is the Purpose of this Application?

The reverse computation research hub serves as a website for researchers and laypeople to come together and learn, read, and discuss topics of reverse computation. Users can share and view publications of pdf format, create posts for events, and share general updates. along with functionality for automatic keyword searching for publications and events. The user can view and sample pseudocode in forward and reverse direction. Previously, alot of this collaboration would be done using primitive methods that limited people in the fields ability to communicate. The goal of this website is to fix that problem.

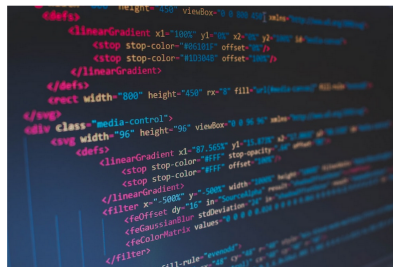
Chapter 2

How to Get Started

The user starts on the homepage, signed out. In this signed-out state, the user can view the homepage, providing a brief introduction to the website and the field of reverse computation. The user can also view publications and events by navigating to their relevant page via the nav bar at the top of the page. A simulator is also available to all users that can run code snippets in an online environment.



Reversible Computation Research Hub



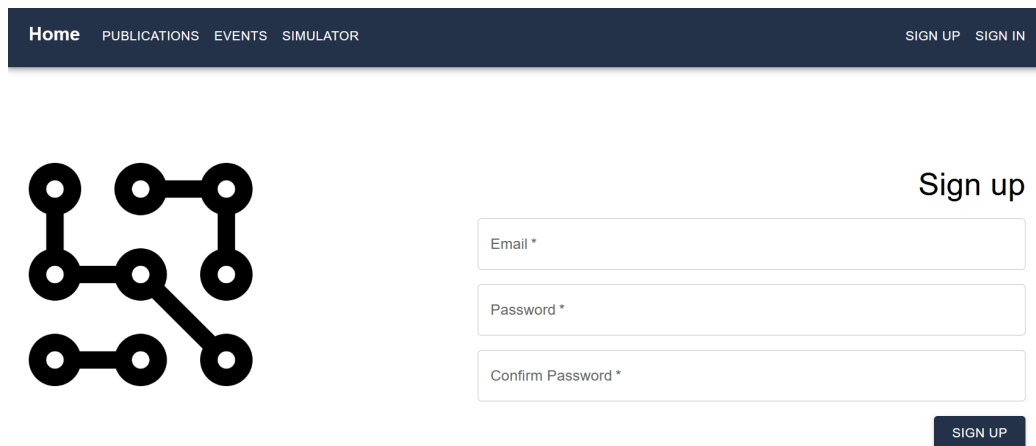
An interesting area of research is reversible computation, namely the ability to begin in a final program state and execute in reverse, correctly restoring each intermediate program state. The community of researchers in this field often collaborate, though without the use of any centralised hub for information sharing.

The aim of this project is to develop a web application that can be the home of reversible computation research. This should allow people to share resources and updates, list the latest publications in this field, as well as upcoming events. The website should also contain a reversible execution simulator, a tool that an interested visitor can use to visualise reversible execution.

Figure 2.1: Homepage for Signed-out Users

2.1 Sign-up

To access full functionality, the user must be signed into an account. If they do not have an existing account they can create one by clicking on the sign-up button on the navigation bar.



Home PUBLICATIONS EVENTS SIMULATOR SIGN UP SIGN IN

Sign up

Email *

Password *

Confirm Password *

SIGN UP

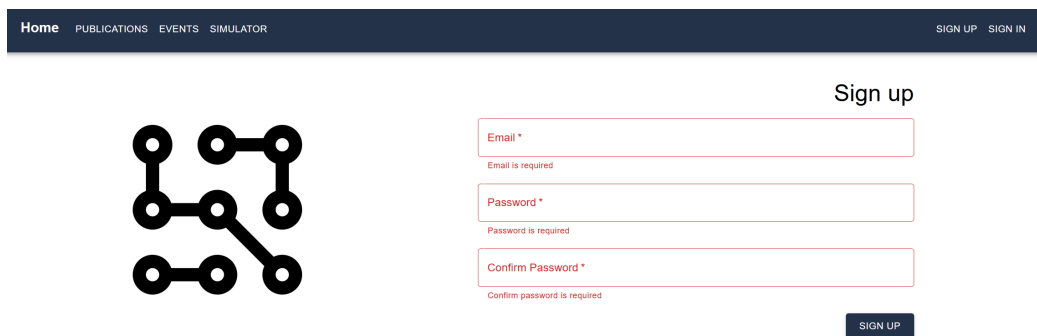
Figure 2.2: Sign-up page

2.2 Errors in Sign-up

If an error occurs with the information inputted, it will be displayed on the webpage. This can then be used by the user to fix the error.

2.2.1 Missing Value

This indicates that there is a value missing in one of the input fields. The fields described will be flagged and can be fixed by adding valid information.



Home PUBLICATIONS EVENTS SIMULATOR SIGN UP SIGN IN

Sign up

Email *

Email is required

Password *

Password is required

Confirm Password *

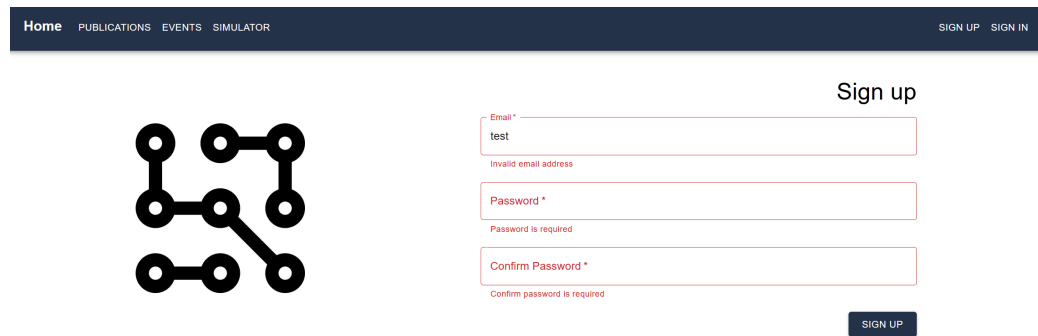
Confirm password is required

SIGN UP

Figure 2.3: Values are missing and need to be entered

2.2.2 Invalid Email

This occurs when the email entered is not a valid email. This can be fixed by ensuring the information inputted is correct.



Home PUBLICATIONS EVENTS SIMULATOR SIGN UP SIGN IN

Sign up

Email *
test
Invalid email address

Password *
Password is required

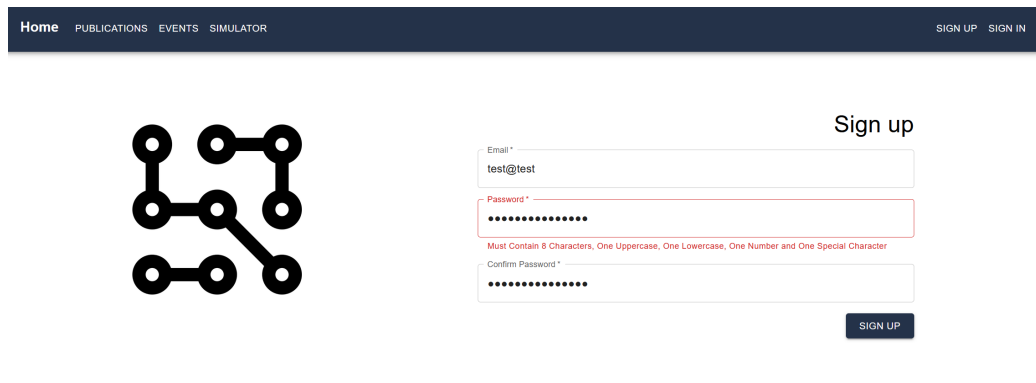
Confirm Password *
Confirm password is required

SIGN UP

Figure 2.4: The email entered is not a valid email and needs changing

2.2.3 Invalid Password

This occurs when the password inputted does not conform to the required character security rules.



Home PUBLICATIONS EVENTS SIMULATOR SIGN UP SIGN IN

Sign up

Email *
test@test

Password *
.....

Must Contain 8 Characters, One Uppercase, One Lowercase, One Number and One Special Character

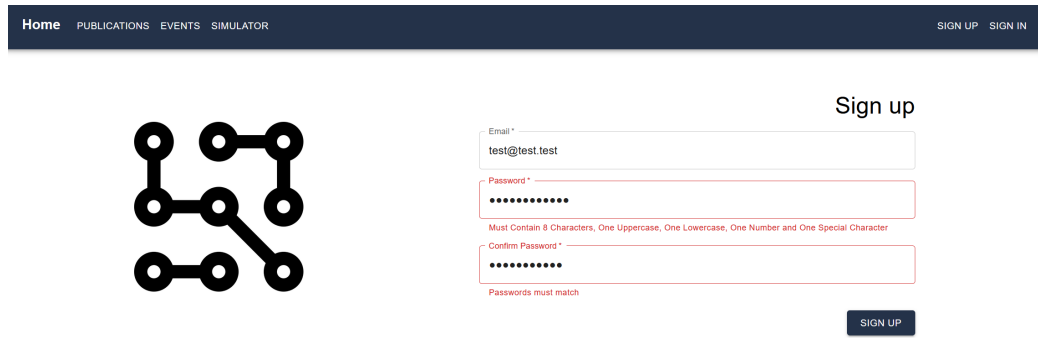
Confirm Password *
.....

SIGN UP

Figure 2.5: Password needs additional characters

2.2.4 Miss-matched Password

This occurs when the passwords entered into the sign-up page do not match. The user should check to ensure both entries of the password match to fix this error.



Home PUBLICATIONS EVENTS SIMULATOR SIGN UP SIGN IN

Sign up

Email *
test@test.test

Password *
.....

Must Contain 8 Characters, One Uppercase, One Lowercase, One Number and One Special Character

Confirm Password *
.....

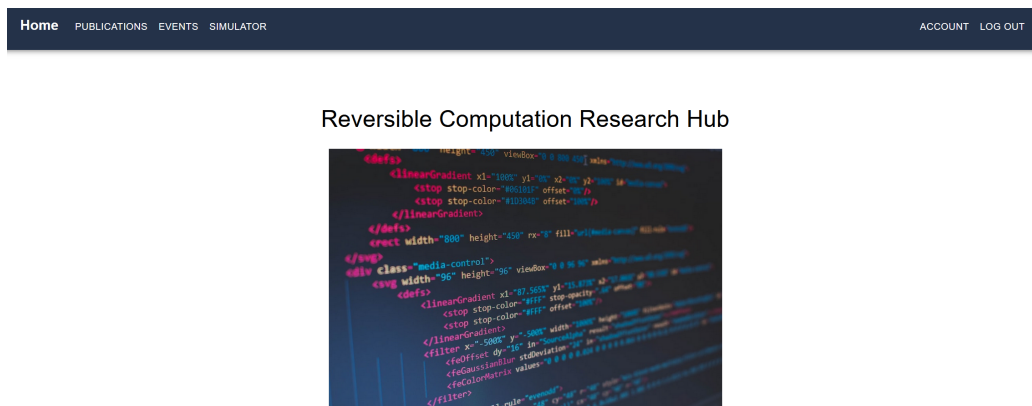
Passwords must match

SIGN UP

Figure 2.6: Passwords do not match

2.3 After Account Creation

The user can enter an email & password to use for their account, and they can then click the 'sign up' button to finish. After this, the user can navigate to the homepage and have access to full functionality. The user can logout to revert to the sign-in page.



An interesting area of research is reversible computation, namely the ability to begin in a final program state and execute in reverse, correctly restoring each intermediate program state. The community of researchers in this field often collaborate, though without the use of any centralised hub for information sharing.

The aim of this project is to develop a web application that can be the home of reversible computation research. This should allow people to share resources and updates, list the latest publications in this field, as well as upcoming events. The website should also contain a reversible execution simulator, a tool that an interested visitor can use to visualise reversible execution.

Figure 2.7: Homepage for logged-in users

2.4 Sign-in

If the user wishes to sign-in using their details, they can now use the sign-in button at the top.



The image shows a sign-in page layout. At the top is a dark blue navigation bar with the text "Group 19" and links for "PUBLICATIONS", "SIGN UP", and "SIGN IN". Below the navigation bar is a large, light gray square area. On the left side of this area is a black icon consisting of nine circles arranged in a 3x3 grid, with lines connecting some of them to form a stylized network or molecule. On the right side of the square area is a sign-in form. The form has a title "Sign in" in bold. It contains two input fields: the first is labeled "Email" and contains the text "hugohul549@gmail.com"; the second is labeled "Password" and contains a series of dots. Below the password field is a dark blue button with the text "SIGN IN" in white.

Figure 2.8: Sign-in page

2.5 Errors in Sign-in

If an error occurs with the information inputted, it will be displayed on the webpage. This can then be used by the user to fix the error.

2.5.1 Value is Required

This indicates that there is a value missing in one of the input fields. The fields described will be flagged and can be fixed by adding valid information.

Group 19 PUBLICATIONS SIGN UP SIGN IN



Sign in

Email

Email is required

Password

Password is required

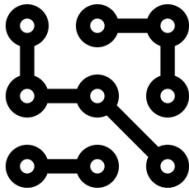
SIGN IN

Figure 2.9: Values are missing and need to be entered

2.5.2 Invalid Email

This occurs when the email entered is not a valid email. This can be fixed by ensuring the information inputted is correct.

Group 19 PUBLICATIONS SIGN UP SIGN IN



Sign in

Email
john.smith

Please enter an email address.

Authentication failed

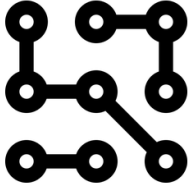
SIGN IN

Figure 2.10: The email entered is not a valid email and needs changing

2.5.3 Authentication Failed

This occurs when the email and password are not found in the user database. This is most likely caused by incorrect information being inputted

Group 19 PUBLICATIONS SIGN UP SIGN IN



Sign in

Emailjohn.smith@domain.com

Password

Authentication failed

SIGN IN

Figure 2.11: Details inputted are not valid credentials

Chapter 3

Account Page

The account page allows the user to access information about their account. At the top of the page is the email that is associated with their account. Below that the user can see their publications and events. These columns also give the user direct access to the edit and delete functions for both publications and events.

- [Editing a publication](#)
- [Deleting a publication](#)
- [Editing an event](#)
- [Deleting an event](#)

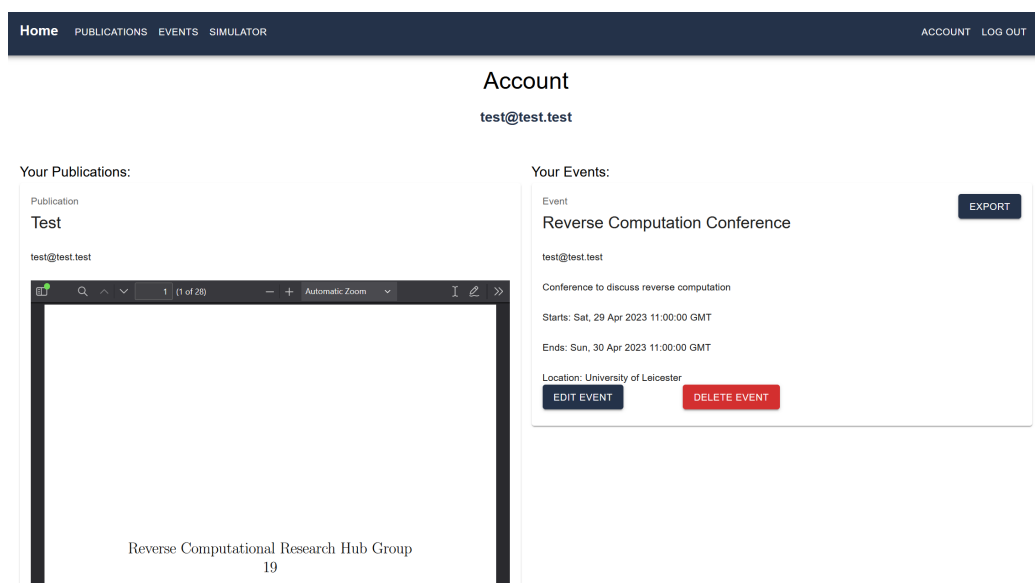


Figure 3.1: Account page displaying account details and posts

Chapter 4

Publications

From the homepage, the user can click ‘Publications’ to be navigated to this section. Here they will see other publication posts with embedded pdfs ordered chronologically along with functionality to create their own publication. A title and a description of the publication is visible.

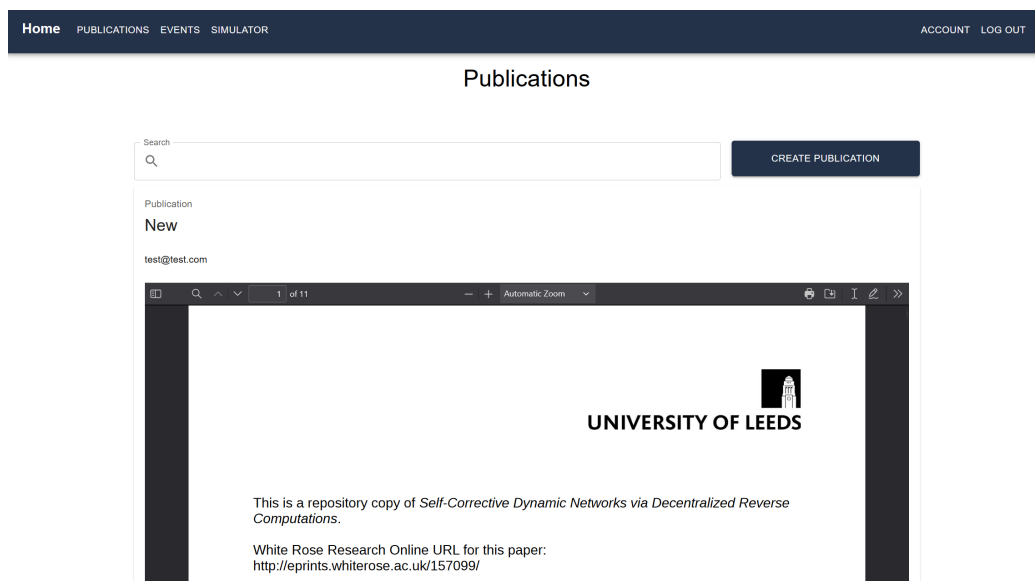
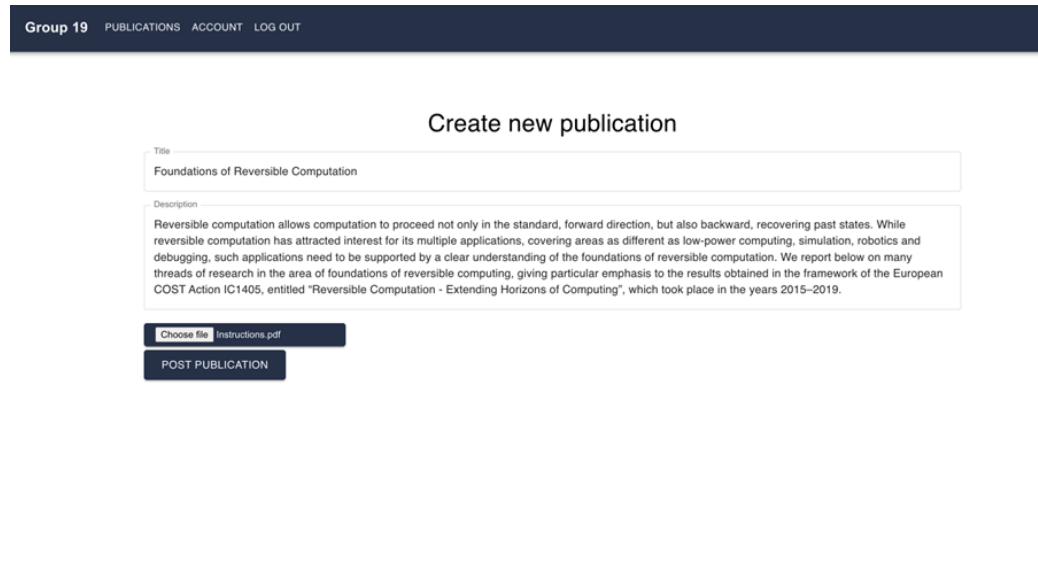


Figure 4.1: Publication page

4.1 Creating a Publication

To create a post, the user can click ‘create publication’.



Group 19 PUBLICATIONS ACCOUNT LOG OUT

Create new publication

Title
Foundations of Reversible Computation

Description
Reversible computation allows computation to proceed not only in the standard, forward direction, but also backward, recovering past states. While reversible computation has attracted interest for its multiple applications, covering areas as different as low-power computing, simulation, robotics and debugging, such applications need to be supported by a clear understanding of the foundations of reversible computation. We report below on many threads of research in the area of foundations of reversible computing, giving particular emphasis to the results obtained in the framework of the European COST Action IC1405, entitled "Reversible Computation - Extending Horizons of Computing", which took place in the years 2015–2019.

Choose file Instructions.pdf

POST PUBLICATION

Figure 4.2: Create New Publication

On this page, the user is provided the opportunity to add a title, description, and a pdf. To upload the pdf, click ‘Choose File’, which will then open the operating system file selector. The user can then click ‘Post Publication’ which then uploads the post to the site. It can then be viewed on the publications page

4.2 Searching for a Publication

At the top of the publications page, the user can search for specific documents. This can be done by entering keywords into the searchbar. The webpage will then update to only show posts that include the keyword.

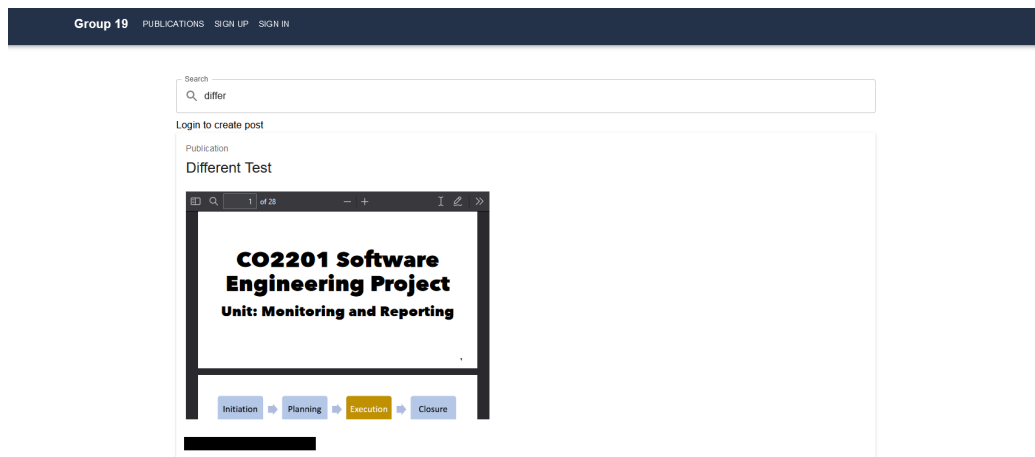


Figure 4.3: Searching the publications for a specific post

4.3 Deleting a Publication

If the user needs to remove a publication for whatever reason, that is also possible. Each post the current user has made has a button which removes the post when clicked.

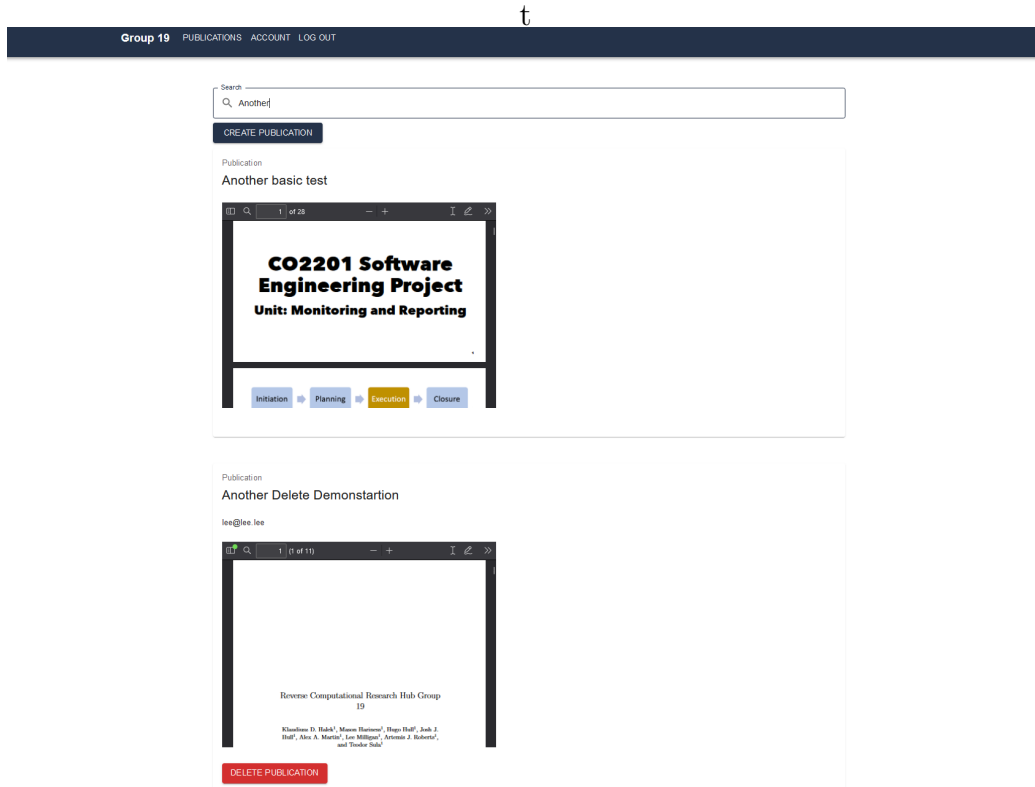


Figure 4.4: The user can only delete posts they made

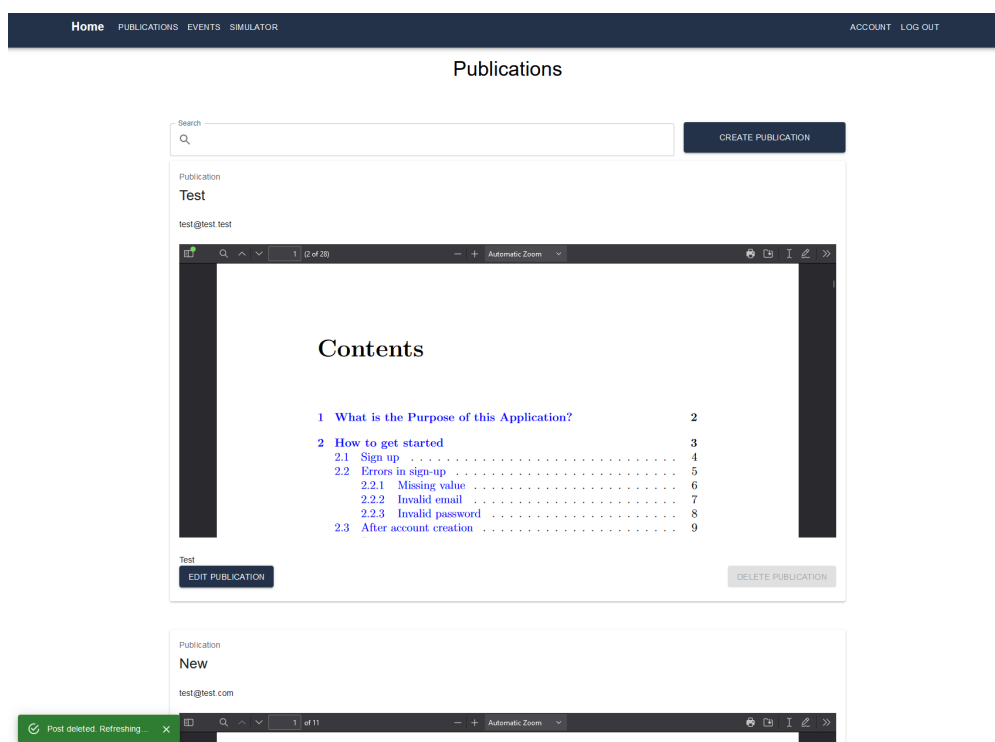


Figure 4.5: The website displays a message to confirm post deletion

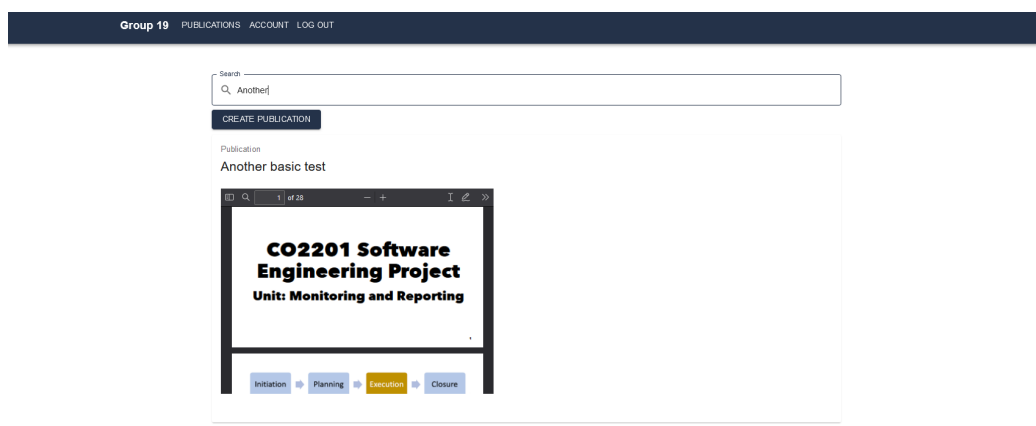


Figure 4.6: The post is no longer found in the page of posts

4.4 Editing a Publication

If the user has inputted incorrect information, they can do so by editing an existing publication. The user can only edit posts they have uploaded. They will only be able to change the title and description of an existing post. If they need to update the uploaded file, they will need to [delete the post](#) and reupload with the correct file.

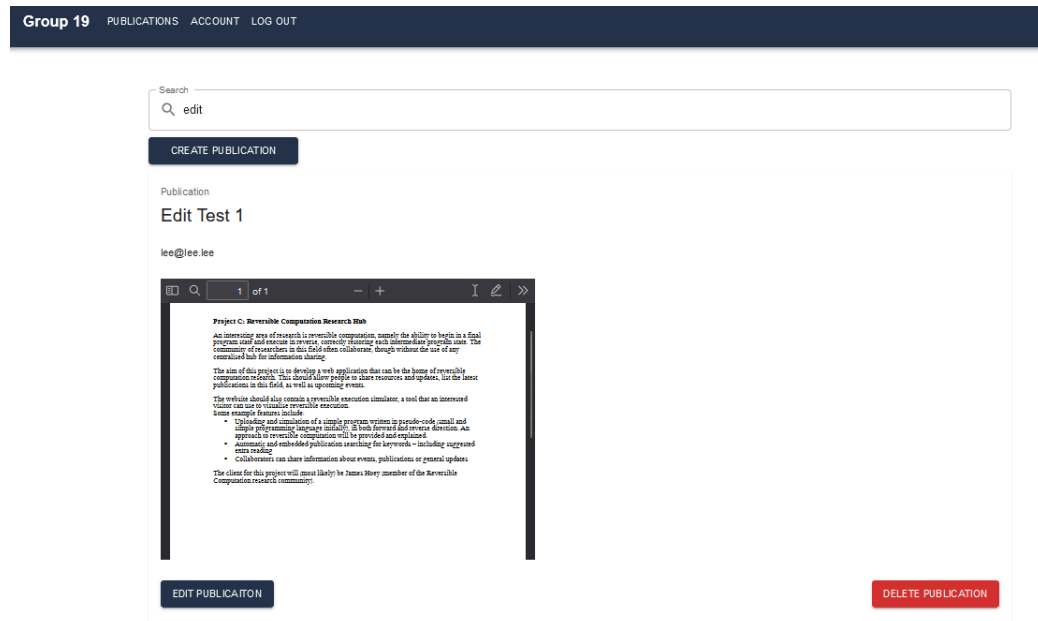


Figure 4.7: Posts you have created have the option to have details about them edited

Group 19 PUBLICATIONS ACCOUNT LOG OUT

Edit publication

Title

Edit Test 1

Description

Edit Test Description

UPDATE PUBLICATION

Figure 4.8: Fields to edit the title and description

Publication

Edit Test 1

lee@lee.lee

1 of 1

Project C: Reversible Computation Research Hub

An interesting area of research is reversible computation, namely the ability to begin in a final program state and execute in reverse, correctly restoring each intermediate program state. The community of researchers in this field often collaborate, though without the aid of any centralized hub for information sharing.

The aim of this project is to develop a web application that can be the home of reversible computation research. This should allow people to share resources and updates, list the latest publications in this field, as well as upcoming events.

The website should also contain a reversible execution simulator, a tool that an interested visitor can use to visualize reversible execution.

Some example features include:

- Uploading and simulation of a simple program written in pseudo-code (small and simple programming language similar to C, but both forward and reverse direction, and approach to reversible computation will be provided and explained).
- Automatic and embedded publication searching for keywords - including suggested extra reading.
- Collaborators can share information about events, publications or general updates.

The client for this project will most likely be James Henry member of the Reversible Computation research community!

Edit Test Description

EDIT PUBLICATION

DELETE PUBLICATION

Figure 4.9: Existing post with new information

Chapter 5

Events

The events page allows users on the website to creating posts with details to real world or online events which other users can then attend. This is helpful to inform the community of conferences and meetups that may be of relevance to them. There is also an area to write out details of the event to get more information across.

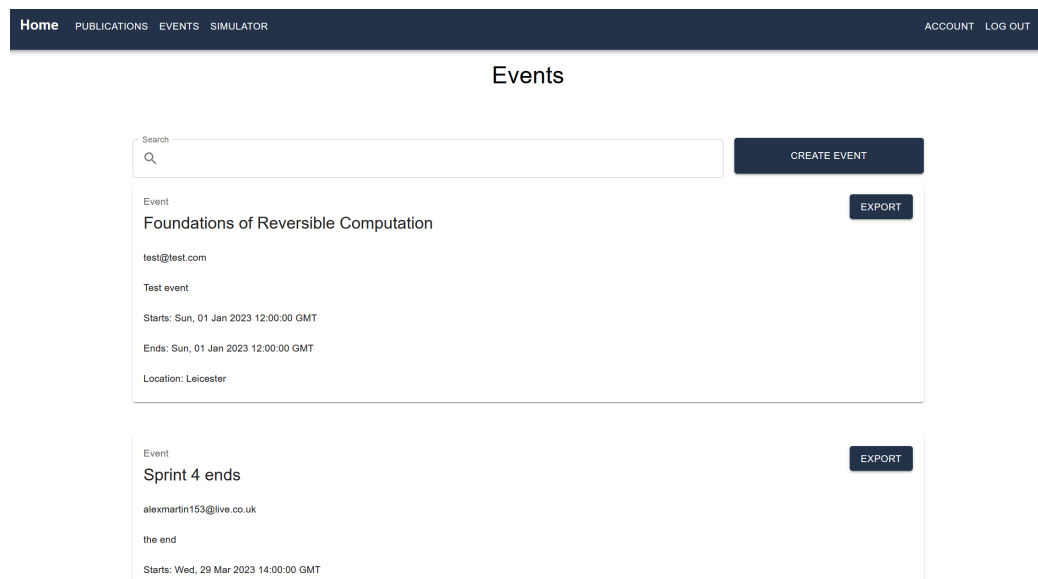
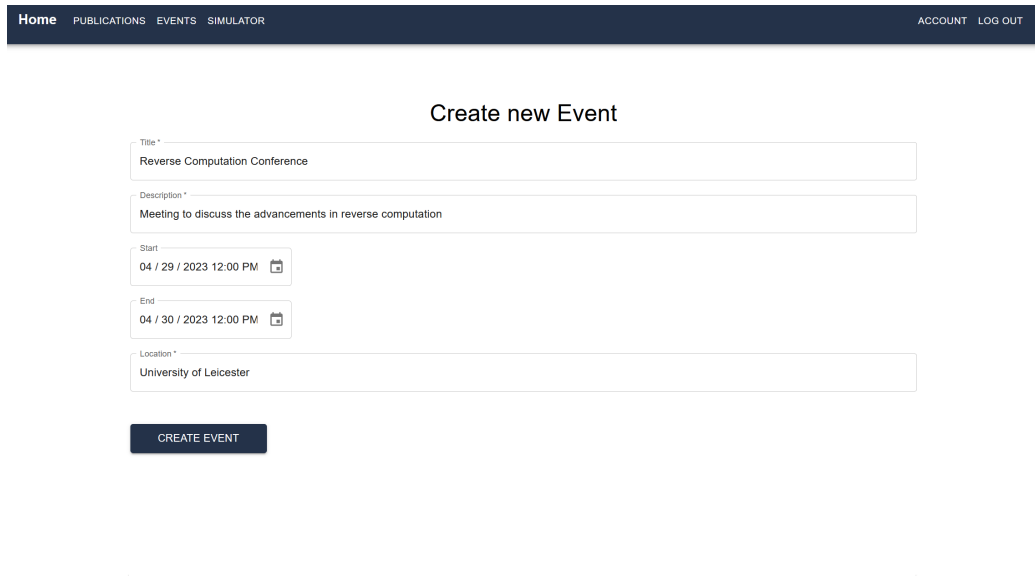


Figure 5.1: Example event page with events listed

5.1 Creating an Event

A logged-in user can create an event by clicking on the 'Create Event' button which can be found next to the event search bar on the main event page. If the user is not logged-in there will be a prompt to login.



The screenshot shows a web application interface for creating a new event. At the top, there is a dark blue navigation bar with links: Home, PUBLICATIONS, EVENTS, SIMULATOR, ACCOUNT, and LOG OUT. Below the navigation bar, the main content area has a heading 'Create new Event'. The form consists of several input fields: 'Title *' with the value 'Reverse Computation Conference', 'Description *' with the value 'Meeting to discuss the advancements in reverse computation', 'Start' with the value '04 / 29 / 2023 12:00 PM' and a calendar icon, 'End' with the value '04 / 30 / 2023 12:00 PM' and a calendar icon, and 'Location *' with the value 'University of Leicester'. At the bottom of the form is a dark blue button labeled 'CREATE EVENT'.

Figure 5.2: Event Creation Page

On the event creation page, there are fields to add: Title, Description, Start time, end time and Location. If the user has made a mistake or need to update the information they originally inputted, they can do so by [editing the event](#).

5.2 Searching the Events for a Specific Listing

At the top of the events page, the user can search for specific events. This can be done by entering keywords into the searchbar. The webpage will then update to only show events that include the keyword.

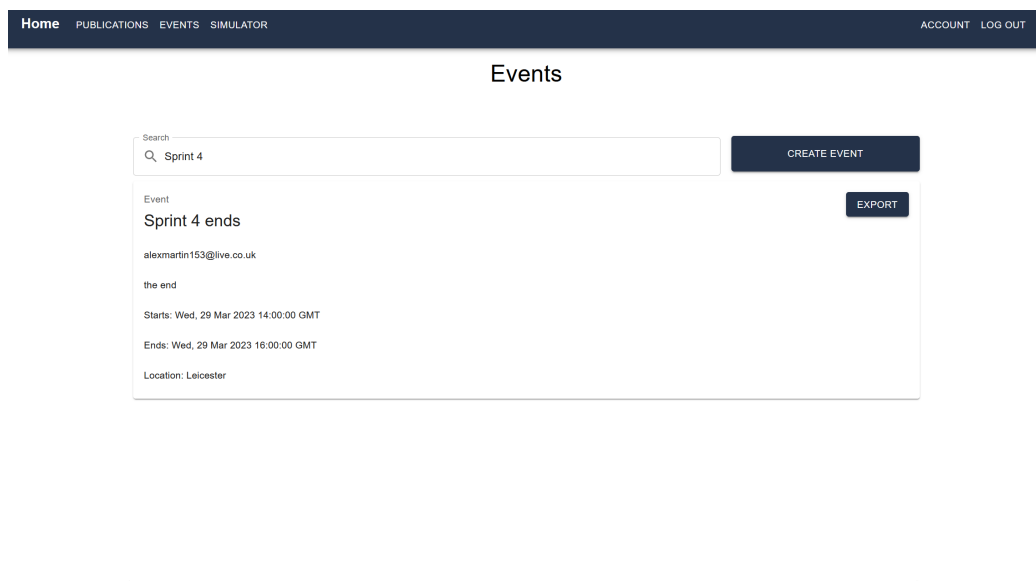


Figure 5.3: Searching for an event

5.3 Deleting an Event

If the user needs to remove an event for whatever reason, that is also possible. Each event the current user has made has a button which removes the post when clicked.

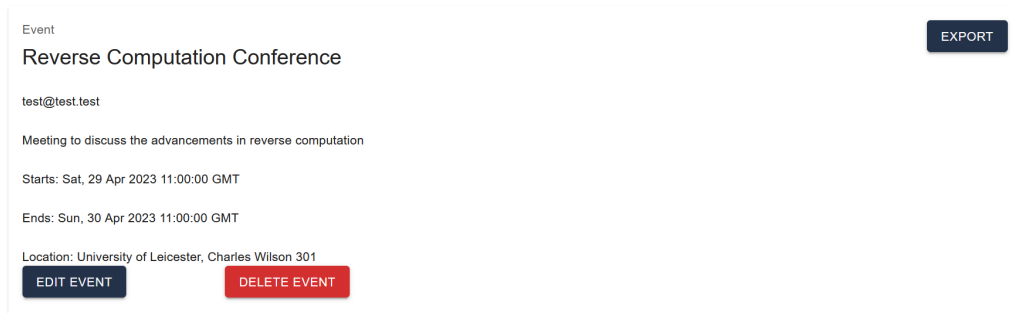


Figure 5.4: Button to delete an event the user created

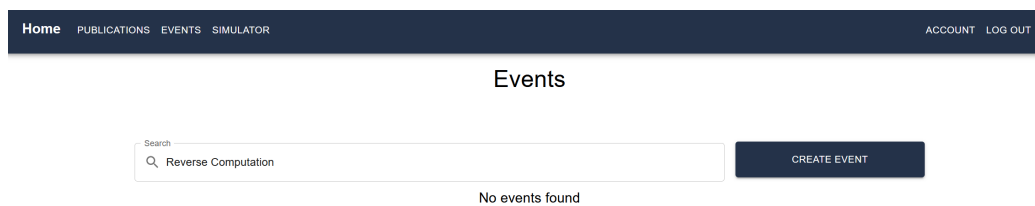
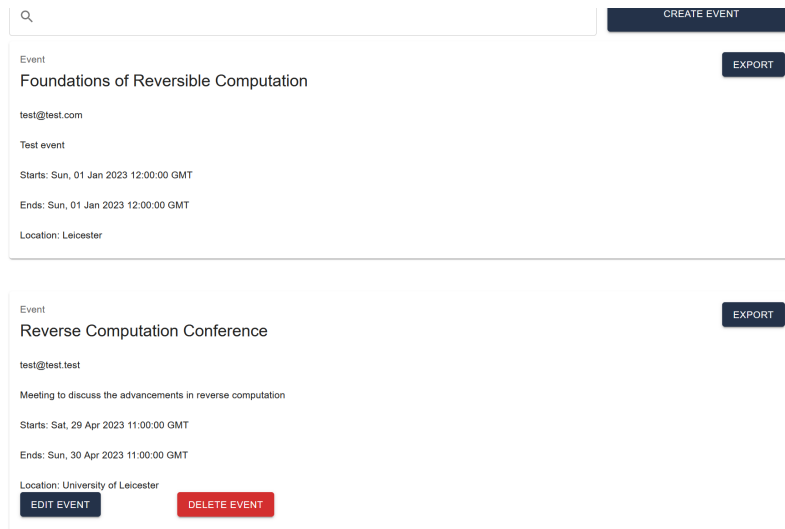


Figure 5.5: Event is deleted from website

5.4 Editing an Event

You are also able to edit events that you have created. This can be done by navigating to your event and clicking the 'Edit Event' button. This will then bring the user to the edit event page.



The screenshot displays a web interface for managing events. At the top, there is a search bar with a magnifying glass icon and a 'CREATE EVENT' button. Below this, two event cards are shown. The first card, titled 'Foundations of Reversible Computation', includes the email 'test@test.com', the label 'Test event', the start time 'Starts: Sun, 01 Jan 2023 12:00:00 GMT', the end time 'Ends: Sun, 01 Jan 2023 12:00:00 GMT', and the location 'Location: Leicester'. It features an 'EXPORT' button. The second card, titled 'Reverse Computation Conference', includes the email 'test@test.test', the description 'Meeting to discuss the advancements in reverse computation', the start time 'Starts: Sat, 29 Apr 2023 11:00:00 GMT', the end time 'Ends: Sun, 30 Apr 2023 11:00:00 GMT', and the location 'Location: University of Leicester'. It features 'EXPORT', 'EDIT EVENT', and 'DELETE EVENT' buttons.

Event	Email	Description	Starts	Ends	Location	Buttons
Foundations of Reversible Computation	test@test.com	Test event	Sun, 01 Jan 2023 12:00:00 GMT	Sun, 01 Jan 2023 12:00:00 GMT	Leicester	EXPORT
Reverse Computation Conference	test@test.test	Meeting to discuss the advancements in reverse computation	Sat, 29 Apr 2023 11:00:00 GMT	Sun, 30 Apr 2023 11:00:00 GMT	University of Leicester	EXPORT, EDIT EVENT, DELETE EVENT

Figure 5.6: Event Before Editing

On the event edit page, you can update any of the fields with new information and then click the 'Edit Event' button to push these changes to the website

Edit Event

Title *
Reverse Computation Conference

Description *
Meeting to discuss the advancements in reverse computation

Start
04 / 29 / 2023 12:00 PM

End
04 / 30 / 2023 12:00 PM

Location *
University of Leicester, Charles Wilson 301

EDIT EVENT

Figure 5.7: Event edit page

Event

Reverse Computation Conference

test@test.test

Meeting to discuss the advancements in reverse computation

Starts: Sat, 29 Apr 2023 11:00:00 GMT

Ends: Sun, 30 Apr 2023 11:00:00 GMT

Location: University of Leicester, Charles Wilson 301

EXPORT

EDIT EVENT DELETE EVENT

Figure 5.8: Event page with updated information

5.5 Exporting Events

For each event, the user has the capability to export the details of the event to either Apple, Google or Outlook calendar. This can be done by the 'Export' button which can be found at the top-right of each event. Once the user clicks it they can choose the calendar of their choice and they will be redirected to the respective page with the details filled in.

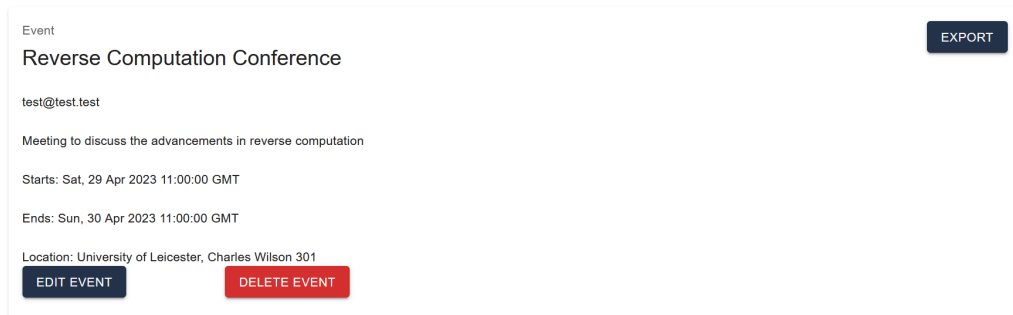


Figure 5.9: Event 'Export' button on the top-right

Chapter 6

Simulator

The simulator provides all users with the ability to discover and learn about reverse computation by putting it into practice. This is achieved by users inputting pseudocode into the simulator. They users can then instruct the simulator to take execute the code step by step in either forward of backwards direction.

6.1 Code Editor

When the user arrives on the simulator page, they are presented with an example set of pseudo code. There is also the option to write their own code if they want.

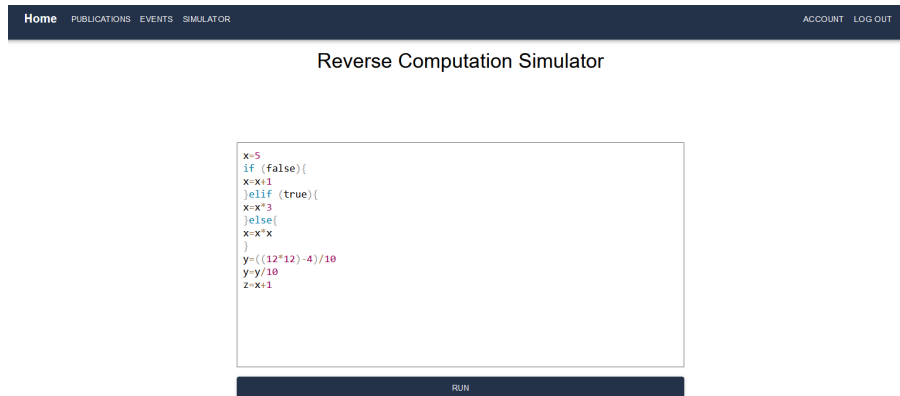


Figure 6.1: Example psuedocode in the editor

6.2 Code Syntax

This section covers the complete syntax guide for the simulator.

6.2.1 Variable Assignment

Variable can be assigned in the code editor by declaring the name of a variable followed by an '=' and then the value you want to assign to the variable. Variable assignment also support arithmetic expressions.

Operator	Name	Example
+	Addition	x=1+1
-	Subtraction	x=2-1
*	Multiplication	x=2*2
/	Division	x=4/2

6.2.2 Comparison Operators

The simulator will support comparing variables against variables and constants.

Operator	Name	Example
==	Equal	x == 2
!=	Not Equal	x != 1
>	Greater then	x >2
<	Less then	x <4
>=	Greater then or equal to	x >= 2
<=	Less than or equal to	x <= 4

6.2.3 Logical Operators

You can use logical operators to perform boolean logic.

Operator	Name	Example
&&	And	x && y
!	Not	!x
——	Or	x —— y

6.2.4 If Statements

The simulator is capable of performing 'If' statements. This allows the program to branch based off conditions that have been set in the program, written by the user. These ifs use the operators mentioned previously.

Operator	Explanation	Example
if	The program will run the branch if the conditions outlined are met	if(false) {x=x+1}
elif	This branch will run if the previous if or elif were not run and its conditions are met	elif(true) {x=x*3}
else	This branch will run if the previous if or elif conditions were not met	else {x=x*x}

6.3 Forward Execution

The user can use the step buttons to execute the code line by line. The webpage will then display the current values for each of the variables that have been declared.

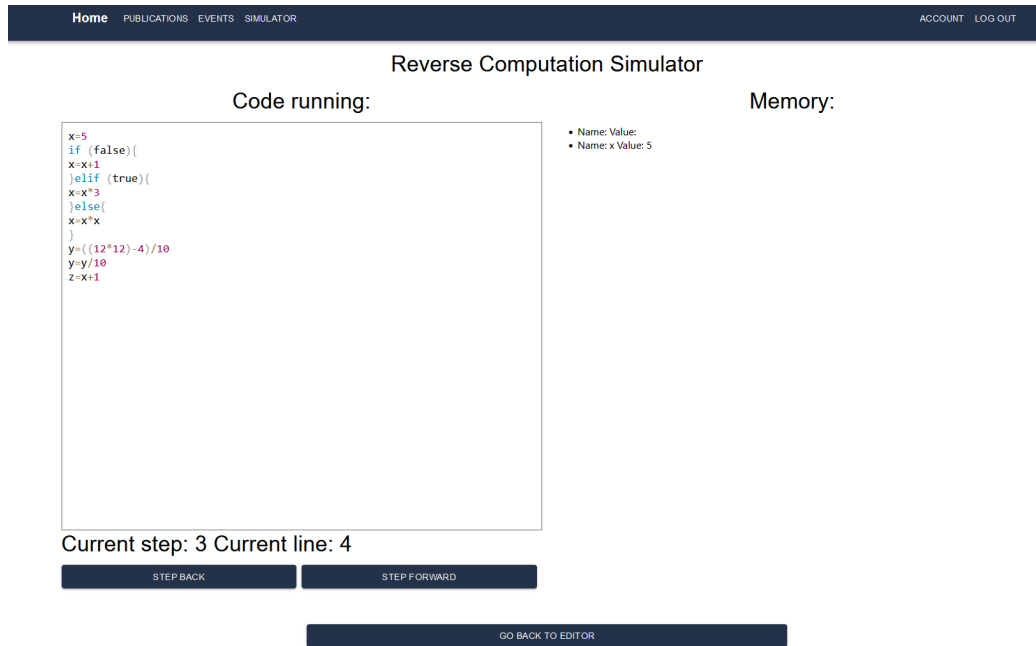


Figure 6.2: Step 3, after three click of 'Step Forward'

[Home](#)
[PUBLICATIONS](#)
[EVENTS](#)
[SIMULATOR](#)

[ACCOUNT](#)
[LOG OUT](#)

Reverse Computation Simulator

Code running:

```

x=5
if (false){
x=x+1
}elseif (true){
x=x*3
}else{
x=x*x
}
y=((12*12)-4)/10
y=y/10
z=x+1

```

Memory:

- Name: Value:
- Name: x Value: 15

Current step: 4 Current line: 5

STEP BACK

STEP FORWARD

GO BACK TO EDITOR

Figure 6.3: Step 4, showing the second line of code has been executed

6.4 Reverse Executiton

The user also has the option to execute the code in reverse. This can be achieved by clicking the 'Step Back' button

[Home](#)
[PUBLICATIONS](#)
[EVENTS](#)
[SIMULATOR](#)

[ACCOUNT](#)
[LOG OUT](#)

Reverse Computation Simulator

Code running:

```

x=5
if (false){
x=x+1
}elseif (true){
x=x*3
}else{
x=x*x
}
y=((12*12)-4)/10
y=y/10
z=x+1

```

Memory:

- Name: Value:
- Name: x Value: 15
- Name: y Value: 1.4

Current step: 8 Current line: 10

STEP BACK

STEP FORWARD

GO BACK TO EDITOR

Figure 6.4: The program is currently at step 8

[Home](#)
[PUBLICATIONS](#)
[EVENTS](#)
[SIMULATOR](#)

[ACCOUNT](#)
[LOG OUT](#)

Reverse Computation Simulator

Code running:

```

x=5
if (false){
x=x+1
}elseif (true){
x=x*3
}else{
x=x*x
}
y=((12*12)-4)/10
y=y/10
z=x+1

```

Memory:

- Name: Value:
- Name: x Value: 15
- Name: y Value: 14

Current step: 7 Current line: 9

STEP BACK

STEP FORWARD

GO BACK TO EDITOR

Figure 6.5: The program steps back to step 7

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2.4	The email entered is not a valid email and needs changing	7
2.5	Password needs additional characters	8
2.6	Passwords do not match	9
2.7	Homepage for logged-in users	10
2.8	Sign-in page	11
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